EEE3097/8/9S Micromouse 2026: Submission 2

GA3 Design Report 1 (20% of Course Mark)

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1. Objective

Submit a formal engineering design report documenting a structured design process conducted in relation to your **closed-loop feedback controller, motor drive, or heading synchronization** from Milestone 1.

The primary purpose is to provide evidence of proficiency in the **ECSA Design Graduate Attribute (GA3)**. This is a mandatory requirement to pass the course.

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2. Format & Submission Rules

To ensure standardization for external ECSA audits, you must adhere to these formatting rules: * **Template:** Populate the Microsoft Word form template located at **docs/assignments/EEE3097_8_9S_designreport.docx**. * **Form Protection: Do not disable the "Protect Form" functionality in MS Word**. The template enforces strict layout limits and character constraints for each section. * **Output:** Save the completed form as a **PDF** and upload it to Gradescope (named `Design_Report_1_[StudentNumber].pdf`).

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3. Open-Ended Design Subsystem Selection

You can choose any design subsystem from Milestone 1 for which you can confidently present evidence of design thinking. Illustrative examples include:

- **Closed-Loop Speed Controller:** Tuning and discretizing a PID feedback loop to match wheel speeds.
- **Robust Heading Alignment:** Fusing gyroscope yaw rate and encoders to steer and correct trajectory drift.
- **DC Motor Drive System:** Modelling motor parameter identification dynamics.

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4. Required Report Sections & Strict Constraints

Strict character constraints (including spaces) are enforced to ensure high-density communication. Populate the template sections detailing: 1. **Brief Description of Design Task (Max 350 chars):** Outline the subsystem task and the problem it solves. 2. **Visual Aids:** Insert a single graphic (e.g. block diagram, Stateflow transition chart, or telemetry plot) referenced in the text. 3. **Design Criteria & Constraints (Max 700 chars):** Define quantitative target metrics and physical/computational limits. 4. **Main Design Assumptions (Max 700 chars):** Outline and mathematically justify your engineering assumptions from first principles. 5. **Design Process Description (Max 700 chars):** Document your structured evaluation of at least two alternative design solutions (e.g., P vs. PID controller). 6. **Design Implementation (Max 700 chars):** Model your chosen design using first-principles dynamic equations or script flow charts. 7. **Design Evaluation & Testing (Max 700 chars):** Verify performance in simulation under physical perturbations (slip, motor asymmetry) and discuss model deviations.